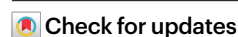


Historical redlining is associated with increasing geographical disparities in bird biodiversity sampling in the United States

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Diego Ellis-Soto¹✉, Melissa Chapman² & Dexter H. Locke³

Historic segregation and inequality are critical to understanding modern environmental conditions. Race-based zoning policies, such as redlining in the United States during the 1930s, are associated with racial inequity and adverse multigenerational socioeconomic levels in income and education, and disparate environmental characteristics including tree canopy cover across urban neighbourhoods. Here we quantify the association between redlining and bird biodiversity sampling density and completeness—two critical metrics of biodiversity knowledge—across 195 cities in the United States. We show that historically redlined neighbourhoods remain the most undersampled urban areas for bird biodiversity today, potentially impacting conservation priorities and propagating urban environmental inequities. The disparity in sampling across redlined neighbourhood grades increased by 35.6% over the past 20 years. We identify specific urban areas in need of increased bird biodiversity sampling and discuss possible strategies for reducing uncertainty and increasing equity of sampling of biodiversity in urban areas. Our findings highlight how human behaviour and past social, economic and political conditions not just segregate our built environment but may also leave a lasting mark on the digital information we have about urban biodiversity.

Biodiversity is increasingly threatened by human activities and rising temperatures with implications for ecosystem functioning and, ultimately, human well-being^{1,2}. The global biodiversity crisis requires immediate action to prevent further population declines and species extinctions². In light of the post-2020 global biodiversity framework on the Convention on Biological Diversity, an improved understanding of the availability and reliability of biodiversity data is critical to equitably and rigorously assess progress towards global and national contributions targets³. Identifying knowledge gaps and guiding efforts towards more equitable sampling efforts are priorities for global biodiversity agendas, such as the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services⁴.

To effectively address the drivers and consequences of biodiversity loss, it is essential to first gather evidence to detect changes in biodiversity over space and through time⁵. The past few decades have seen a drastic growth in the amount of available biodiversity data, driven primarily by technological advances and digital collection from citizen scientists using cellphone apps, such as the eBird platform to collect bird observations⁶. However, data collected by volunteers are highly spatially biased with species records' locations⁷. Consequently, these large datasets may have gaps or redundancies that complicate subsequent analyses⁸. The lack of data in certain geographic regions, strong sampling bias and preference towards specific taxa will limit the inferences on where and why species live within and across regions^{9–11}.

¹Department of Ecology and Evolutionary Biology, Yale University, New Haven, CT, USA. ²Department of Environmental Science, Policy, and Management, University of California, Berkeley, CA, USA. ³Baltimore Field Station, Northern Research Station, USDA Forest Service, Baltimore, MD, USA.

✉e-mail: diego.ellisoto@yale.edu

Research on so-called ‘biodiversity knowledge gaps’, therefore, tends to explore coarse patterns (that is, comparing countries) of biodiversity across the world and commonly identifies the tropics as data poor and the northern hemisphere as data rich^{9,10}. Overall, there appears to be socioeconomic, linguistic and ecological drivers for these gaps, and biases in our current understanding and available information on biodiversity¹⁰. For example, sampling bias in biological fossil records is driven by legacies of colonialism and political stability, which hinders a more representative view of our past biodiversity¹².

Urban biodiversity knowledge might be particularly sensitive to the social, economic and cultural environments of cities: most urban biodiversity data come from citizen scientists whose movements through urban landscapes are influenced, if not guided, by these factors¹³. Urban environmental injustice patterns might be worse than previously documented due to biased sampling campaigns by volunteer efforts. This is particularly relevant as bird observations from citizen-science campaigns, such as eBird and the iNaturalist platforms, are already actively being used for a large array of conservation outcomes^{14,15}, such as creating conservation targets for migratory birds^{16,17}. As a consequence, resulting data gaps could lead to conservation actions limited to only certain parts of cities. The problem may be particularly acute in the United States because of large socioeconomic inequality¹³.

Histories of and present-day socioeconomic inequalities may lead to highly disparate sampling of biodiversity records across neighbourhoods. A study using the citizen-science eBird platform found that higher-income areas of Raleigh (North Carolina) and Tucson (Arizona) were overrepresented and lower-income areas were undersampled by volunteers¹⁸. However, there were no differences in sampling density by income in Durham (North Carolina) and Fresno (California), highlighting the importance of cross-site data in addressing questions of volunteered geographic information’s social and spatial biases in an urban biodiversity context¹⁸. This is because environmental amenities are unevenly distributed, especially in urban areas, with lower-income areas having lower tree canopy cover¹⁹ and access to parks²⁰. Spatially biased biodiversity sampling in urban areas represents a pivotal limitation for equitable and just planning of sustainable cities, as we risk underestimating urban biodiversity in undersampled, low-income neighbourhoods. How can we protect what we don’t even know we have?

In the United States, racial segregation is the norm, not the exception²¹. One notable race-based housing policy colloquially referred to as ‘Redlining’ in the 1930’s categorized neighbourhoods by racial composition and housing stock, and then used those maps to guide mortgage loans. As part of The New Deal, the Home Owners’ Loan Corporation (HOLC) appraised neighbourhoods on the basis of their perceived safety of real estate investment using a ranking system of A (green), B (blue), C (yellow) or D (red, hence ‘redlining’). A consequence of differential loan access is blocking the transfer of wealth intergenerationally. Today, across the United States, lower-graded sections exhibited declines in home ownership²² and home values, higher vacancy rates²³ and higher poverty²⁴. Historic redlining practices have also become associated with shorter life expectancies and poorer overall health in present day²⁵, and lower educational achievement²⁶.

There are also environmental correlates of redlining. Studies of more than 100 metropolitan regions show that areas formerly inhabited by marginalized groups have currently less vegetation²⁷ overall and are hotter²⁸ than neighbourhoods formerly consisting of US-born, white residents. A study using high-resolution, high-accuracy tree canopy maps for 37 US cities found that neighbourhoods composed predominantly of US-born white neighbourhoods in the 1930s have on average 43% tree canopy today, whereas areas with formerly predominantly marginalized groups have on average 23% tree canopy cover today¹⁹. Street tree species diversity is lower in formerly D-graded areas when compared with their A-graded counterparts in Baltimore, Maryland²⁹. The current distributions of greenspaces and biodiversity are influenced both by current and past land uses³⁰. Birds are without

doubt the best sampled taxa and constitute approximately 64% of the globally available biodiversity information (based on gbif.org queried on 4 May 2023) collected in most parts by hundreds of thousands of citizen scientists using the eBird citizen-science platform³¹.

A growing body of evidence on the relationships between systemic racism, socioeconomic disparities and environmental conditions across urban areas further allows making assumptions on how these social conditions relate to the ecology and evolution of urban biodiversity^{32,33}. Volunteered avian biodiversity data have been shown to vary by median household income in some cities but not others¹⁸. Specifically, how biodiversity varies with the degree of historic residential segregation within and among cities appears to be a fundamental knowledge gap to our present-day understanding and distribution of biodiversity³². Investigating these hypotheses is intended to guide future environmental justice research in relation to biodiversity science.

This paper explores how historic zoning policies relate to bird data collection across and within 195 cities in the United States after controlling for greenspace and climate (Fig. 1). Specifically, we look at the key building blocks for reliable biodiversity estimates: sampling density of biodiversity records to identify potential biases and differences in data collection and survey completeness which indicates certainty in documenting all bird species in a region or whether more sampling is necessary to make reliable inferences. We ask (1) how does sampling effort of bird biodiversity vary with socioeconomic conditions (historic housing segregation policies, such as HOLC-graded neighbourhoods), biophysical predictors across urban environments and by data source; (2) whether the least-sampled areas for bird biodiversity (biodiversity coldspots hereafter) are consistently located in regions that were segregated by redlining; and (3) whether when considering temporal trends, biodiversity sampling is becoming more even or uneven over time in the age of increasing digital and citizen-science data collection?

Results

Sampling density and biodiversity coldspots

We found differences in sampling across HOLC grades. A total of 8% of HOLC A neighbourhoods did not display any bird records compared with 13% for grade-D polygons. Formerly A-graded areas have more than twice the sampling density (1,592 per km²) of formerly D-graded areas (712 samples per km²) (Fig. 2a). Beyond sampling density, how complete our understanding of bird biodiversity is across HOLC grades was also significantly different (Wilcoxon rank-sum $W = 886,278$, $P = 0.009$), with redlined areas being the least-complete areas of study (Fig. 2b), and C- and D-graded areas having the largest number of biodiversity coldspots (Fig. 2c).

Predictors of bird sampling density

Summary statistics of the predictors are shown in Supplementary Table 1. A mixed-effects model showed that the adjusted odds of being sampled at all are 36% lower (odds ratio (OR): 0.64; 95% confidence interval (CI): 0.49, 0.83; $P < 0.001$) in D than in A neighbourhoods (Table 1). Another mixed-effects model revealed that sampling density varied significantly across all HOLC grades, when adjusting present-day vegetation greenness (NDVI), open space (the percentage of the polygon covered by protected open space) and population density (people per km²; Table 1 and Supplementary Table 3). Climate did not appear to be a meaningful predictor after performing model selection (Supplementary Table 2). NDVI, open space and population density were retained as significant predictors. Sampling density was greatest in formerly A-graded neighbourhoods. Sampling density odds are 37% lower in B-graded neighbourhoods (OR: 0.63; 95% CI: 0.54, 0.73; $P < 0.001$); 63% lower in C-graded neighbourhoods (OR: 0.37; 95% CI: 0.31, 0.43; $P < 0.001$) and 74% lower in D-graded neighbourhoods (OR: 0.26; 95% CI: 0.21, 0.31; $P < 0.001$) (Fig. 3). Overall, the rank-order of the HOLC grading system matches the sampling density of bird sampling efforts. However, some cities deviate from that pattern (Supplementary Fig. 1).

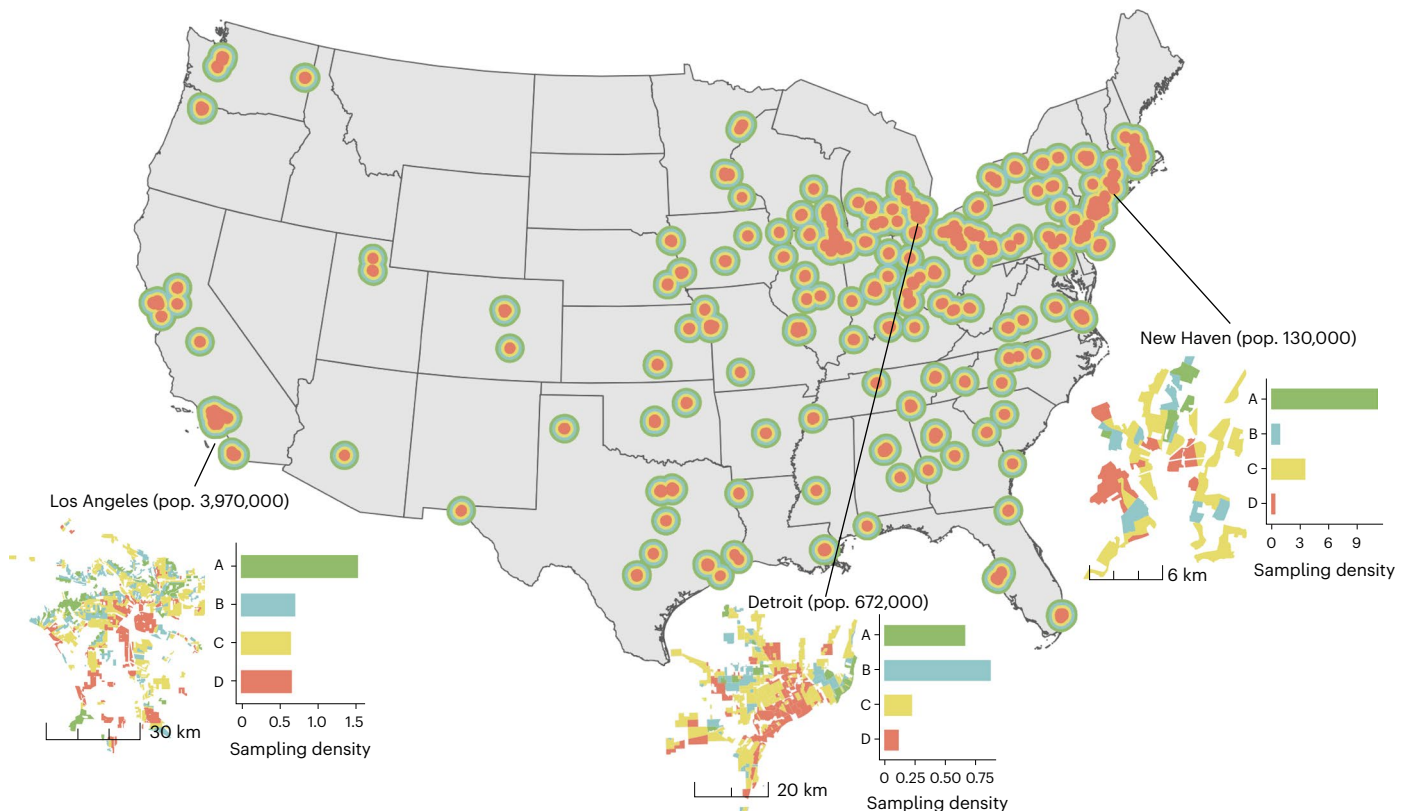


Fig. 1 | Map of 195 selected cities included in the analyses, spanning a range of social and ecological conditions. New Haven (Connecticut), Los Angeles (California) and Detroit (Michigan) show how formerly D-graded (or ‘redlined’) areas have far lower citizen-science sampling density for birds.

Bird survey completeness

A third mixed-effect model of survey completeness also showed significant variation by HOLC grade when adjusting for vegetation and open space, although not all pairs of grades were different (Table 1). Completeness was greatest in formerly A-graded (-65.39% ; 95% CI: 63.41, 67.36) and lowest in formerly D-graded areas (-59% ; $\beta = -7.10$; 95% CI: $-9.11, -5.08$; $P < 0.001$) at mean NDVI, percent open space and population density values (Fig. 3). This suggests greater uncertainty for estimates of key biodiversity metrics, such as species richness and survey completeness, in grade-D areas. Recognizing that Type I and Type II sum of squares analysis of the mixed-model’s outputs can lead to different inferences, Type I and Type II results are provided in Supplementary Table 3. These tables reinforce the findings from analyses offered in Table 1 and Fig. 3 because they lead to the same inference, namely, that there are differences in the outcome (sampling density or completeness) by HOLC grade when controlling for the covariates.

Bird biodiversity data through time and by observation type

Our generalized additive model revealed that year and HOLC grade were both significant predictors in explaining the cumulative sampling of bird biodiversity through time ($P < 0.001$; Supplementary Table 4). More specifically, sampling of bird biodiversity appears to be becoming more uneven through time, with gaps across HOLC grades expanding in recent years (2000–2020), driven by the explosion in biodiversity sampling, particularly by the eBird citizen-science platform (Fig. 4 and Supplementary Fig. 2). The ratio of sampling density between HOLC A and D grades was 2.021 in the year 2000 and grew to 2.74 by 2020, representing an increase in sampling disparity of 35.6%.

The most common citizen-science platforms, eBird and iNaturalist, appear to express different biases in sampling density, with eBird being more unevenly distributed, having more observations in HOLC A and B than in HOLC C and D grades (Fig. 3 and Supplementary Fig. 2).

While iNaturalist data were much more evenly distributed across HOLC grades, it constitutes only a small fraction of the data ($\sim 1\%$) compared with eBird (Supplementary Fig. 2). Further, data collected by the Global Biodiversity Information Facility (GBIF), such as museum records and data collected by academics, which we refer to as ‘other’ in terms of bird biodiversity data type, were unevenly distributed across HOLC grade, following a trend similar to that of eBird, with HOLC A (8.73; 95% CI: 6.51, 11.7; $P < 0.01$) and B (9.32; 95% CI: 7.30–11.9; $P < 0.01$) having higher sampling density compared with HOLC C (6.06; 95% CI: 4.83, 7.62; $P < 0.01$) and D (6.06; 95% CI: 4.83, 7.62; $P < 0.01$) grades (Fig. 2a).

Discussion

Our study shows disparities in the density of bird records from the past ~ 90 years in historically marginalized communities across 195 cities in the United States (Fig. 2). The odds of sampling in formerly D-graded areas is 74% lower, and the survey completeness is 7.10 percentage points lower (or $\sim 1\%$ lower) than formerly A-graded neighbourhoods (Table 1). Taken together, this evidence highlights how formerly redlined areas are significantly undersampled compared with non-redlined areas and that there is overall less certainty about the number of bird species that are known to occupy these areas. These differences persisted even after accounting for differences in vegetation, open space, population density and climate (Supplementary Table 2), and appear to be getting more uneven with time (Fig. 4 and Supplementary Table 4). This trend appears particularly marked in eBird, the largest citizen-science initiative for bird data collection in the world (Supplementary Fig. 2).

These data disparities caution against comparing bird biodiversity within cities and across urban neighbourhoods given that sampling intensity is the first building block for biodiversity estimates, unless further corrections are taken³⁴. If samples are not representative, they are biased, as we see across HOLC grades. This can dramatically

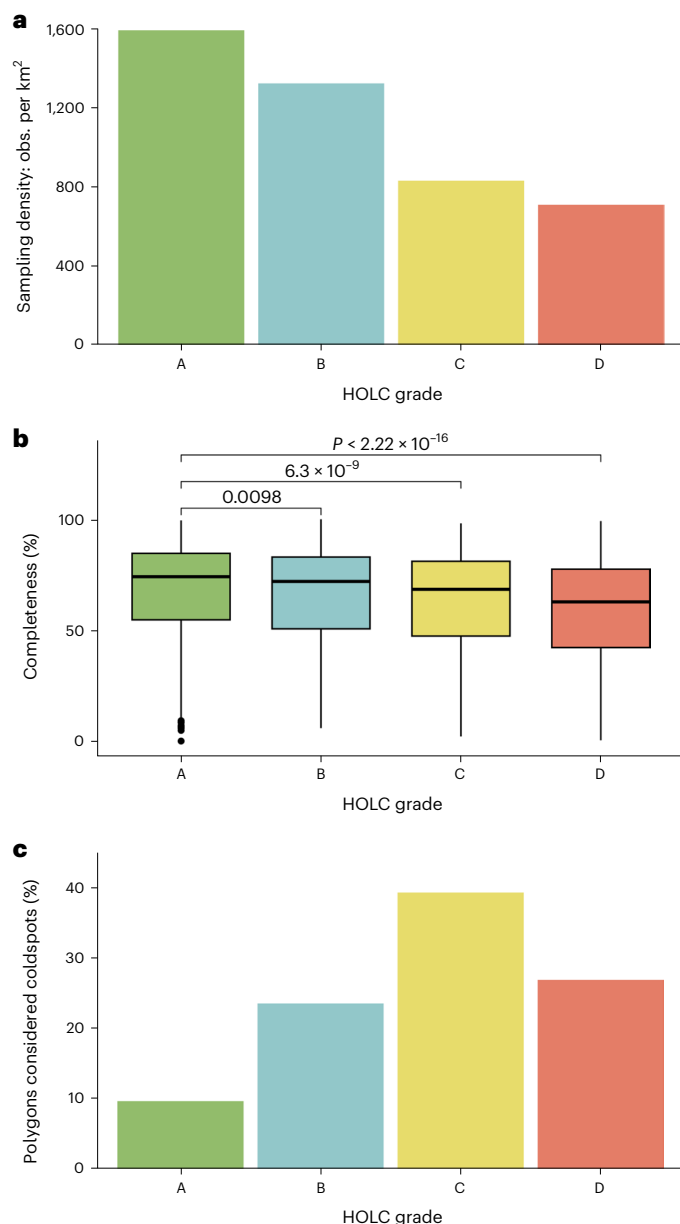


Fig. 2 | Patterns of biodiversity across redlined areas. a, Sampling density (bird records per km²) is lowest in HOLC D neighbourhoods. **b,** The completeness of available bird biodiversity information differs significantly across HOLC grades, with significantly lower completeness in redlined, D-graded areas (pairwise, two-sided Mann–Whitney *U* test). Medians are denoted by horizontal lines, boxes represent the interquartile range (IQR) and whiskers are 1.5× the IQR. **c,** The largest proportion of neighbourhood polygons deemed insufficiently sampled (coldspots) in terms of bird biodiversity are found in HOLC C grades, followed by D grade.

influence measurements of biodiversity change³⁵, hindering our ability to accurately map and forecast species' distributions. Urban bird biodiversity knowledge appears biased towards areas that used to be single-family detached homes occupied by US-born, white families—formerly A-graded areas. However, this does not necessarily mean that these areas have higher biodiversity. A limitation of our study is not accounting for current demographics across redlined neighbourhoods in addition to using relatively coarse urban greenness measures. Future studies may focus on individual cities using local high-resolution canopy cover for a more in-depth exploration of how environmental amenities might relate to sampling of bird biodiversity.

Ecological research increasingly includes social-ecological processes³⁶, yet incorporation of environmental justice remains low³⁷. Environmental justice highlights the need for a just and even access and sampling of natural resources, including opportunities to sample biodiversity, as well as a fair representation in education and policy³⁷. This raises the question of how to address environmental injustice scholarship within the most widely adopted citizen-science activity: bird watching. Without an even sampling of natural resources, understanding of ecosystem services will remain limited in marginalized communities, hindering environmental sustainability for the most vulnerable groups. We found that the legacy of redlining is negatively associated with our understanding of current urban bird biodiversity, limiting our ability to accurately map bird species distributions in cities. Moreover, this trend appears to be worsening over time despite our ability to collect bird biodiversity information digitally. This might represent an environmental justice of concern, since the disparity will probably grow in the future if left unaddressed. In responding to the thought-provoking questions of Schell and colleagues³², we find that biodiversity sampling varies by former residential segregation via HOLC grades.

Demographic disparities in participation in citizen science have been documented extensively, including in bird watching^{38,39}. Citizen science is often predominantly collected by adult, well-educated, white and affluent users^{38,39} who can be highly specialized birders⁴⁰. Simultaneously, sites frequently sampled by citizen-science efforts may be in areas of low environmental justice concern³⁸, which can in turn lead to disparate representation of sampling (such as shown in our study). Previous work has suggested that to reduce disparities, citizen-science initiatives could 'centre in the margins', meaning that if existing projects are accessible to the marginalized, they will be accessible to all⁴¹. Aligning scientific research goals of improving sampling of bird biodiversity in areas identified as coldspots in our analysis, along with local grassroots organizations (non-governmental organizations, neighbourhood associations) involved in increasing scientific literacy (that is, STEM high-school programmes using citizen-science applications, Black Birders Week (<https://www.audubon.org/black-birders-week>, <https://www.blackafinstem.com>)) or promoting nature walks for marginalized groups (such as Latino Outdoors <https://latinooutdoors.org>), could be a step towards more-inclusive citizen science and serving groups historically and currently underserved. In addition, participating citizen scientists may benefit from increased environmental advocacy^{41,42}, a closer connection to nature, increased scientific literacy of public research participants through relationships with academic institutions and communities⁴³, and physical activity in urban greenspaces indirectly promoting physical and mental health benefits⁴⁴.

Local communities harnessing their power in data collection may better advocate environmental protections in their own community⁴⁵. More importantly, a lack of representation and information about biodiversity may lead to uneven funding for environmental restoration and benefits to communities, thus reinforcing environmental justice disparities⁴⁶. Such a feedback loop may promote a cycle of continued disinvestment in areas of underrepresented and underserved populations and their environments. We find, however, that these areas are where the greatest knowledge gaps exist and the greatest uncertainty about species' spatial distribution and abundance remains. Ultimately, citizen science is a volunteer activity and merits the question whether participants are seeking to fill important gaps in our urban knowledge of birds, or for purely recreational purposes, or some combination of both^{40,47}. The coldspots map identifies opportunities to optimize the value of information volunteers collect for improved scientific inference while simultaneously advancing environmental justice (Supplementary Data 1).

A better understanding of urban biodiversity is beneficial both for local communities residing in the area and for improving accuracy, representativeness and understanding of biodiversity overall,

Table 1 | Mixed-effects models predicting whether a HOLC neighbourhood is sampled (left), the sampling density (middle) and sampling completeness (right)

Sample binary				log(sampling density)			Completeness		
Predictors	Odds ratios	CI	P	Odds ratios	CI	P	Estimates	CI	P
(Intercept)	12.85	9.50–17.38	<0.001	239.57	202.27–283.73	<0.001	65.39	63.41–67.36	<0.001
HOLC grade B	0.93	0.71–1.22	0.585	0.63	0.54–0.73	<0.001	–1.72	–3.51–0.07	0.060
HOLC grade C	0.94	0.72–1.21	0.614	0.37	0.31–0.43	<0.001	–3.69	–5.47–1.90	<0.001
HOLC grade D	0.64	0.49–0.83	0.001	0.26	0.21–0.31	<0.001	–7.10	–9.11–5.08	<0.001
NDVI				0.93	0.88–0.99	0.024	1.11	0.36–1.87	0.004
pct pa				1.45	1.39–1.51	<0.001	2.73	2.18–3.28	<0.001
pop. per km ²				1.33	1.26–1.40	<0.001	2.05	1.45–2.64	<0.001
Random effects									
σ^2	3.29	3.73	478.77						
τ_{00}	1.24 msa_NAME	0.41 msa_NAME	43.15 msa_NAME						
τ_{11}		0.10 msa_NAME.holc_gradeB							
		0.14 msa_NAME.holc_gradeC							
		0.30 msa_NAME.holc_gradeD							
ρ_{01}		1.00							
		0.61							
		0.12							
ICC	0.27		0.08						
N	143 msa_NAME	143 msa_NAME	142 msa_NAME						

The P values are a simple approximation, based on the t-statistic and using the normal distribution function. $P < 0.05$ are bolded. pct pa, percent protected area; pop., population; σ^2 , residual variance; τ_{00} , intercept variance or random intercept variance; τ_{11} , slope variance or random slope variance; ρ_{01} , random slope-intercept correlation; ICC, intraclass correlation coefficient (a.k.a. variance partition coefficient) which is the proportion of the variance explained by the grouping structure in the population; N, number of observations included in the model.

especially as it relates to sustainability and climate goals. We envision that biodiversity sampling and coverage becomes included as an environmental justice index by the US Environmental Protection Agency (EPA; <https://www.epa.gov/ejscreen>)—something that is currently lacking, particularly for key species such as birds or insect pollinators. The federal environmental justice initiative by the current Biden Administration, labelled the Justice40 Initiative, seeks to ensure that at least 40% of federal climate and energy programme funding to communities is directed to communities that are and have in the past been most affected by environmental injustice (<https://www.whitehouse.gov/omb/briefing-room/2021/07/20/the-path-to-achieving-justice40/>). The administration aims to use environmental justice screening tools by the EPA to identify disadvantaged communities for this benefit programme. This executive order also aims to track performance towards the Justice40 Initiative by assisting vulnerable communities as well as benefitting these communities with infrastructure investment. By accounting for bias in biodiversity sampling through a novel EPA indicator, future work could identify neighbourhoods requiring improved and more just biodiversity sampling; the Justice40 Initiative may allow the mobilization of federal funding to empower local communities by providing K12 STEM educational opportunities, which may benefit historically marginalized communities while improving our understanding of urban bird biodiversity.

At a global scale, the 2030 United Nations Sustainable Goal 11 aims to achieve recognition for sustainable and inclusive urban growth that

minimizes inequality in global policy agendas⁴⁸. Thus, understanding the effect of socio-ecological dynamics on environmental justice is a crucial knowledge gap⁴⁹. Besides redlining, there are several policies across the world that have segregated communities and led to socio-economic and environmental inequalities. This highlights the potential of building upon this research to uncover uneven distribution of biodiversity information through an environmental justice lens in other areas of the world. Increasingly, people's experience with nature will occur in cities, with potential benefits for human health and well-being linked to biodiversity conservation⁵⁰. Centering justice in urban bird monitoring through whole-community engagement and mobilization could substantially increase our understanding of biodiversity in identified coldspots⁵¹ and provide a foundation towards a more environmentally just sampling of biodiversity across urban areas.

Methods

HOLC polygons

The HOLC polygons were obtained for a total of 195 cities from the University of Richmond website⁵². These maps were digitized and georeferenced as part of the Mapping Inequality Project (<https://dsl.richmond.edu/panorama/redlining/>) and as HOLC-defined neighbourhoods for each city. We calculated the total geographic area of each HOLC grade for each city in km² using the sf package (v.1.0.12)⁵³ in R (v.4.1.1)⁵⁴ and the tigris package (v.1.5) to obtain county-level information for each city⁵⁵. Population estimates for each HOLC polygon were created via

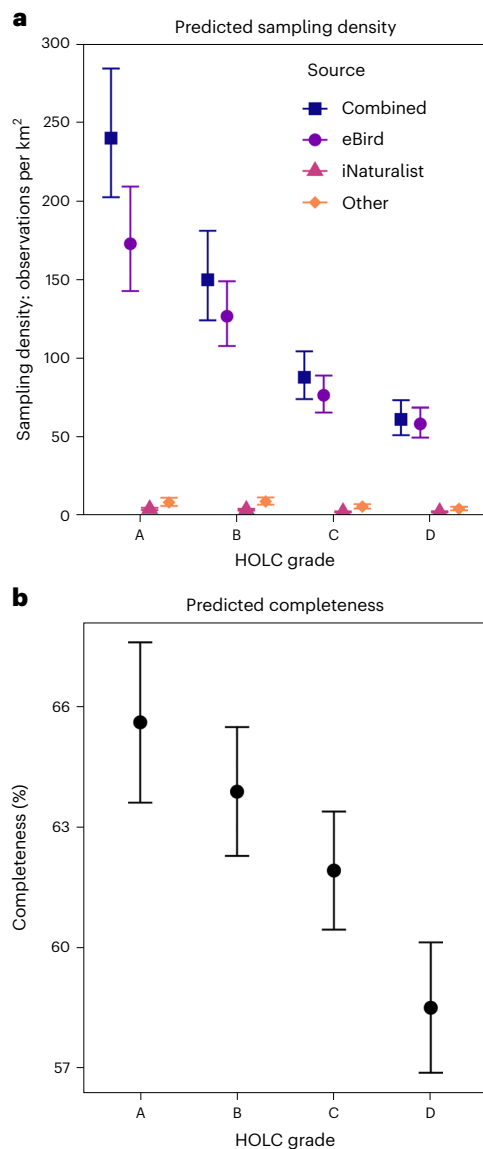


Fig. 3 | Predictions of sampling density and completeness. a, Predicted sampling density by data source adjusted for NDVI, open space (%) and population density (population per km²), with random intercepts per MSA and random slopes for HOLC grade in that MSA. **b,** Predicted percent completeness adjusted for NDVI, open space (%), population density and random intercepts for MSA. Note that the y axis does not start at zero. Error bars represent 95% confidence intervals.

areal interpolation, whereby each neighbourhood was assigned the weighted sum of the proportional overlap of each Census block group from the 2013–2017 American Community Survey population using spatial intersection⁵⁶.

Bird biodiversity records

We obtained a total of 12,297,506 georeferenced biodiversity records for bird species (class Aves) occurring within 9,851 HOLC-defined neighbourhoods in 195 cities from the GBIF (<https://gbif.org>), which also contains records from the eBird database³⁷. We used the rgbif package (v.3.7.3.1)⁵⁸ to query and download bird records that had geographic coordinates and were collected after 1932 (as the HOLC began categorizing neighbourhoods in 1933) up to October 2022. We included records defined as ‘observation’, ‘living specimen’, ‘human observation’, ‘preserved specimen’ and ‘machine observation’. We calculated the

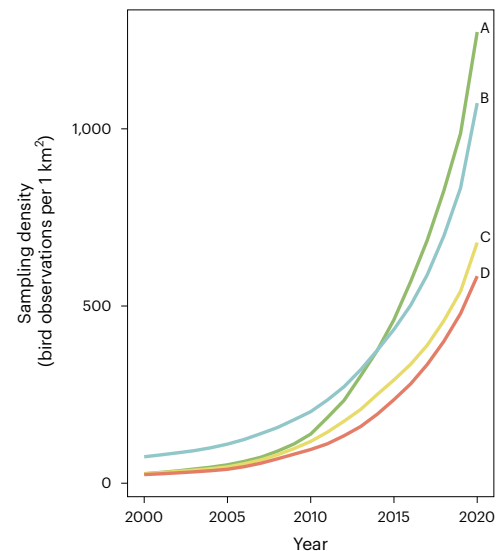


Fig. 4 | Temporal trends in cumulative sampling density of bird observations in km² across HOLC grades from 2000 to 2020. In the age of digital biodiversity data, there is a rapid increase in the amount of biodiversity data collected, with differences in sampling density increasing and becoming more uneven over time. Sampling density is highest in HOLC A neighbourhoods and lowest in HOLC D neighbourhoods, with a sampling disparity of ~40% by 2020 when compared to 2000.

number of bird observations as well as the species richness (number of unique bird species) for each HOLC grade for each city. To calculate how the number of bird observations differed across HOLC grades, we calculated the sampling density of bird records by accounting for the area of each HOLC grade per city in km² and log transformed it to meet model assumptions. After removing HOLC polygons that contained no bird records, we retained a total of 8,909 HOLC polygons in 195 cities for further analysis. Because extensive work has demonstrated the importance of greenspace for bird presence in urban environments^{59,60}, we modelled sampling density separately on the basis of the area of greenspace habitats for each HOLC grade in cities by using the Protected Area Database of the United States (PAD-US) (described below).

To investigate temporal trends in bird data collection and the type of records collected, we calculated the yearly number of bird records first from 1933–2022. Then, we filtered our observations from 2000–2020 to investigate the role of digital information in shaping the distribution of records across HOLC grade by observation type. We used the collection code and institution code column from our GBIF records to calculate sampling density for eBird, iNaturalist and other datasets separately. The category ‘other’ included all non-eBird and -iNaturalist datasets, and includes records such as museum collections.

Calculating bird sampling gaps and survey completeness across HOLC in cities

To investigate disparate survey efforts for bird biodiversity across HOLC grade in cities, we used the KnowBR package (v.2.0) to calculate survey completeness of bird information in a geographic unit; individual HOLC polygons in our case⁶¹. Specifically, survey completeness calculated by KnowBR reflects our ability to quantify the full species assemblage in a HOLC polygon, defined as the percentage of observed species richness captured by expected species richness⁶¹. Two cities, San Antonio and Houston, had to be excluded from survey completeness analysis as their polygon geometries did not allow calculations using KnowBR.

Calculating bird information coldspots

We further used KnowBR to estimate incomplete sampling of bird information (‘coldspots’ hereafter) across redlined areas. Besides

survey completeness, we calculated the slope of the species accumulation curves as well as the completeness percentage for each HOLC polygon. Specifically, we used the reversed rationale used in refs. 59,61 to remove poorly surveyed areas. Thus, we considered polygons to be incompletely sampled (bird information coldspots) using parameter recommendations from ref. 61: the slope of the species accumulation curve is >0.3 , survey completeness is below 50% and the ratio between occurrence record numbers and the number of observed species as <3 . This allowed us to calculate the percentage of incompletely sampled redlined polygons. These areas could be used to guide local residents, ornithologists and citizen scientists for an overall more-spatially-even sampling of bird biodiversity.

Present-day vegetation, open space and climate data

We used the PAD-US⁶² to calculate the percent area of parks within each neighbourhood. The PAD-US includes parcels with small spatial extents, thus incorporating greenspaces that may not be identified by coarser remote-sensing products. While incomplete, the PAD-US data represent shared open space in urban areas. Layers from the PAD-US have been successfully used to explore patterns of bird biodiversity using eBird in the past⁶³. In addition, we created a composite NDVI map with Google Earth Engine, using an atmospherically corrected surface reflectance from the Landsat 8 sensors. We took all images taken during a 5-year period (2014–2019), masked clouds and calculated mean NDVI per month using the normalized difference calculated as (near infrared – red) / (near infrared + red). This results in a number between 1 and –1, where pixels with high photosynthetic activity have a high NDVI. We then calculated the mean value per redlining polygon. NDVI captures a distinct process from the PAD-US database in urban areas: NDVI represents vegetation patterns, which includes trees in residential and private parcels of land; PAD-US, by contrast, represents shared open space. We obtained the mean annual precipitation and mean annual temperature built on monthly averages of climate data collected from meteorological stations around the globe from 1979 to 2013 at 1-km² resolution⁶⁴ and calculated average temperature and rainfall for each city.

Statistics

Three mixed-effects models were estimated. The first estimated whether a HOLC neighbourhood was sampled at all (binary outcome) using a generalized linear mixed model with lme4's glmer function⁶⁵, the second estimated the log of sampling density and the third estimated survey completeness using linear mixed models with lme4's lmer function⁶⁵. We assessed model performance on the basis of Akaike information criterion minimization selection between simple and complex models with an increasing number of covariates⁶⁶. The simplest null model (intercept only) was fit and fixed effects for HOLC (primary variable of interest) were added. Next, the following terms were added: metropolitan statistical area (MSA) as a random intercept, HOLC grade as a random slope, and fixed effects for NDVI, percent open space and population density as random slopes within HOLC-defined city for mean temperature and precipitation (climate) interaction. Because NDVI, percent open space and population density were measured on very different scales, they were standardized via z-scores before being used as predictors. For the binary outcome (sampled or not) and log of sampling density, the random intercept and random slope model was best, and for survey completeness, random intercept. In addition to providing model predictions, Type I analysis of variance was computed on the lme4 objects with the base R ANOVA function. The primary variable of interest, HOLC grade, was entered into the mixed model as the last fixed effect. To assess the robustness of our findings to the order of variables, we computed both Type I and Type II sums of squares analysis of variance using the ANOVA function base package in R⁶⁶. To estimate whether biodiversity sampling is becoming more even or uneven over time, we calculated a generalized additive model

predicting the yearly bird sampling density as a function of HOLC grade and year with an interaction effect.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data used in this analysis are publicly available on Zenodo (<https://doi.org/10.5281/zenodo.8052525>) and were queried from GBIF as a derived dataset (<https://doi.org/10.15468/dd.ha9ksv>). Climatic predictors used in our analysis are publicly available through the Chelsa Climatology platform (<https://chelsa-climate.org/>), remote-sensing data are available on the Google Earth Engine (<https://earthengine.google.com>) and the PAD-US dataset can be downloaded from the United States Geological Survey website (<https://www.usgs.gov>).

Code availability

The codes used to download biodiversity data and environmental data, as well as for necessary analysis and creation of figures to recreate our results are available on Zenodo (<https://doi.org/10.5281/zenodo.8052525>).

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Author contributions

D.E.-S., M.C. and D.H.L. designed the study and performed research. D.E.-S. led the writing of the paper with substantial contributions from M.C. and D.H.L.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Diego Ellis-Soto.

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Data collection A collection of code used for data collection and analysis is available on Zenodo (<https://zenodo.org/record/8052525>) as well as on GitHub (https://github.com/DHLocke/HOLC_uneven_biodiv).

Data analysis All analysis were run using R v. 4.1.1. and the packages tigris v. 1.5, sf v. 1.0.12, rgbf v. 3.7.3.1 and KnowBR 2.0.

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We obtained biodiversity records from the Global Biodiversity Information Facility (<https://gbif.org>) which contains data from the eBird database (<https://ebird.org/>). Biodiversity data used in this analysis is publicly available on Zenodo (<https://zenodo.org/record/8052525>) and was queried as a derived dataset (<https://doi.org/10.15468/dd.ha9kqv>). Redlining maps were downloaded from the Mapping Inequality Project (<https://dsl.richmond.edu/panorama/redlining/>).

Climatic predictors used in our analysis are publicly available through the Chelsa Climatology platform (<https://chelsa-climate.org/>) and remote sensing data is available on Google Earth Engine (<https://earthengine.google.com>) and the PAD-US dataset can be downloaded from the United States Geological Survey website (<https://www.usgs.gov/>)

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Study description	Historic segregation and inequality are critical to understanding modern environmental conditions. Race-based zoning policies, such as redlining in the United States during the 1930s, are associated with racial inequity and adverse multigenerational socioeconomic levels in income and education, and disparate environmental characteristics including tree canopy cover across urban neighborhoods. We quantify the association between redlining and bird biodiversity sampling density and completeness - two critical metrics of biodiversity knowledge - across 195 cities in the United States. We show that historically redlined neighborhoods remain the most under-sampled urban areas for bird biodiversity today, potentially impacting conservation priorities and propagating urban environmental inequities. The disparity in sampling across redlined neighborhoods grades increased by approximately 40% over the past 20 years. We identify specific areas critical for reducing uncertainty and increasing equity of sampling of biodiversity in urban areas. Our findings highlight how human behavior and past social, economic and political conditions segregate not just our built environment, but may also leave a lasting mark also the on the digital information we have about urban biodiversity.
Research sample	We obtained a total of 12,297,506 georeferenced biodiversity records for bird species (class Aves) occurring within 9,851 HOLC-defined neighborhoods in 195 cities, from the Global Biodiversity Information Facility (https://gbif.org), which also contains records from the eBird database 58. We used the rgbif package v. 3.7.3.1 59 to query and download bird records that had geographic coordinates, and were collected after 1932 (as the Home Owners Loan Corporation began categorizing neighborhoods in 1933) up to October 2022. We included records defined as 'observation', 'living specimen', 'human observation', 'preserved specimen' and 'machine observation'.
Sampling strategy	As we were interested in the sampling of bird biodiversity and the amount of records per km2, we downloaded all bird records occurring within our defined HOLC neighborhoods in 195 cities collected between 1933 and 2022.
Data collection	We obtained bird biodiversity records from the Global Biodiversity Information Facility (GBIF) using the 'rgbif' package version 3.7.3.1 in R v. 4.1.1.
Timing and spatial scale	We queried and downloaded bird records that had geographic coordinates intersecting with our HOLC polygons and were collected from 1933 up to 2022.
Data exclusions	After downloading biodiversity records defined as 'observation', 'living specimen', 'human observation', 'machine observation' from the Global Biodiversity Information Facility (GBIF), we did not exclude any datasets
Reproducibility	Data and code used in this analysis is publicly available on Zenodo repository (https://zenodo.org/record/8052525) as well as on GitHub (https://github.com/DHLocke/HOLC_uneven_biodiv)
Randomization	As we were interested in the sampling of biodiversity records, we did not randomize any samples
Blinding	Blinding was not used for data acquisition in this manuscript as we downloaded more than a dozen million bird records for our study across and within 195 metropolitan areas in the United States across 38 states

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